WEEK 4 — Cognitive Aspects (Explained like a story)

Imagine you're designing an app and your user is a human brain. The brain is powerful, but it has limits:

- It can't pay attention to everything at once.
- It sees patterns, but also gets tricked.
- It remembers better with cues than without cues.
- It prefers learning by doing, not by reading manuals.
- Under stress (or multitasking) it makes more errors.
  So if your interface ignores these limits, people struggle — and then blame themselves.
  HCI says: don't blame the user. Design for how cognition actually works.

The lecture covers:

1. What cognition is + why it matters in HCI
2. Core cognitive processes: attention, perception, memory, learning, reading/speaking/listening, problem-solving
3. Cognitive frameworks: mental models, gulfs of execution/evaluation, distributed cognition, external & embodied cognition

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1) What is cognition?

Definition

Cognition = mental activities like **thinking, remembering, learning, decision-making, seeing, reading, talking, writing**.

Two big "high-level" ways to classify cognition

A) Experiential vs Reflective (Norman, 1993)

- **Experiential cognition**: "in the moment," fast reacting, skilled actions, effortless (e.g., driving on a familiar road).
- **Reflective cognition**: slow thinking, comparing, deciding, creating new ideas (e.g., planning a trip).

B) Fast vs Slow Thinking (Kahneman, 2011)

- **Fast thinking (System 1)**: automatic, intuitive, little effort (e.g., 2+2, eye color).
- **Slow thinking (System 2)**: conscious effort, mental work (e.g., 21×29, recalling first school name).

Why cognition matters for HCI

Understanding cognition helps because it:

1. tells what users can/can't be expected to do
2. explains why users face problems
3. provides theories and tools to design better products

MEMORIZE (teacher nitpick):

- Definition of cognition + examples
- Experiential vs reflective + fast vs slow
- 3 reasons cognition is important in HCI

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2) Core cognitive processes (and design implications)

2.1 Attention

What is attention?

Attention = selecting what to focus on from many stimuli. It allows focus, but limits tracking everything.

Types:

- **Focused attention**: one thing
- **Divided attention**: multitasking

Important study: layout affects attention (Tullis, 1987)

Two screens had same info density (31%) but different layout:

- Cluttered screen: avg **5.5s** search
- Grouped with spacing & vertical categories: avg **3.2s** search

Why? Spacing and grouping guide attention; clutter forces scanning chaos.

Multitasking and attention

Multitasking causes losing train of thought, errors, restarting. Heavy multitaskers are *more distractible*, struggle filtering irrelevant info.

Core reason: multitasking overloads limited focus capacity.

Attention in critical context: driving + phone

Phone use while driving increases reaction time because cognitive resources split; even hands-free is not “safe” because distraction is cognitive. Passenger is less dangerous than remote caller because passenger shares context and pauses when hazard appears.

Design implications for attention (MEMORIZE)

- Make key info **salient** when needed: color, ordering, spacing, animation
- Avoid clutter
- Support switching and returning to task

2.2 Perception

Perception = how sensory info becomes experience. Design must ensure representations are easily perceivable: legible text, distinguishable icons.

Example idea from activities:

- Borders + spacing grouping often helps faster search than color contrast alone (Weller, 2004).

Design implications (MEMORIZE list):

- Icons should be clearly distinguishable
- Use borders/spacing to group info
- Sounds should be audible & distinguishable
- Use proper color contrast (yellow on white/green is bad)
- Use haptic feedback carefully

2.3 Memory

Memory = encoding + retrieval of knowledge for action. We don’t store everything; attention affects encoding; context affects retrieval. Recognition is easier than recall.

Working memory vs long-term memory (MEMORIZE)

- **Working memory**: small amount, recent (e.g., last sentence in conversation)
- **Long-term memory**: older, large store (e.g., tune from years ago)

Context matters

Out-of-context retrieval is harder (neighbor on train example).

Recognition vs Recall (very testable)

- **Recall**: retrieve without cues (command-line: remember command names)
- **Recognition**: choose from visible options (menus/icons/tabs/history lists)

Key exam line: Humans are much better at **recognition** than **recall**, so GUIs are easier.

The “7±2” problem (misapplication)

Miller’s 7±2 is about recalling arbitrary chunks, but in interfaces users often scan/recognize, so menus can have more than 7 items depending on task and screen space.

Personal Information Management (PIM)

We store tons of files; naming is common but hard to remember. Design can help with metadata: timestamps, tags, categories, and search (e.g., Spotlight).

Memory design implications (MEMORIZE)

- Reduce cognitive load (avoid complex procedures)
- Design for recognition, not recall (golden rule)
- Provide labeling/organization methods: folders, tags, colors, flags, timestamps

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2.4 Learning

Learning = accumulation of skills/knowledge. Two types:

- **Incidental learning**: effortless (faces, what happened today)
- **Intentional learning**: effortful (studying) — harder

People prefer **learning by doing**, not manuals.

Design implications (MEMORIZE):

- Encourage exploration
- Constrain/guide learners (don't let them get lost)
- Link concepts and representations dynamically for complex learning

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2.5 Reading, speaking, listening

People vary:

- Some prefer listening; listening can require less cognitive effort than reading/speaking
- Dyslexics face difficulty with written words

Applications:

- Voice UIs (Siri/Alexa)
- Speech output (text-to-speech for visually impaired)
- Chatbots (natural language text)

Design implications (MEMORIZE):

- Speech menus/instructions should be short
- Artificial speech is harder to understand → use better intonation
- Offer large text options

2.6 Problem-solving, planning, reasoning, decision-making

These are reflective cognition: weighing options, consequences, sometimes using artifacts (maps, paper), scenarios, alternatives.

Design implications (MEMORIZE):

- Provide easy access help/info for deeper understanding
- Use simple, memorable functions to support quick decision-making/planning

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3) Cognitive frameworks (how we explain/predict user behavior)

3.1 Mental Models (VERY IMPORTANT)

Definition (MEMORIZE)

Mental model = user's internal understanding of how a system works:

- what to do next
- how the system behaves in unfamiliar situations
  Users make inferences using these models.

Craik (1943): mental models are internal constructions that allow predictions; can be shallow or deep.

Thermostat example (classic)

Many people think: "set thermostat to max → heats faster" (general valve theory "more is more"). But thermostat is basically on/off; setting higher doesn't change heating rate. This is an **erroneous mental model**.

Other erroneous mental model examples:

- pressing elevator button repeatedly (believing it speeds it up)

How design helps build correct mental models (MEMORIZE)

- clear instructions
- tutorials + contextual guidance
- videos/chatbots
- transparency + affordances (what actions are possible)

3.2 Gulfs of execution and evaluation (Norman)

The "gulfs" are gaps between user and system.

- **Gulf of execution**: gap between user goal and actions the system allows ("What can I do? How do I do it?")

- **Gulf of evaluation**: gap between system state and user understanding ("What happened? Did it work?")

Bridging gulfs reduces cognitive effort: make actions obvious + give clear feedback.

MEMORIZE: definitions + the question each gulf answers.

3.3 Distributed cognition

Cognition is not only in one person's head; it's distributed across:

- people + artifacts + environment
  Information propagates across displays, paper, speech, tools.

Example: airplane cockpit / air traffic control — safe design requires understanding the whole cognitive system.

MEMORIZE: "across individuals, artifacts, representations."

3.4 External cognition + offloading

External cognition = how we use external representations (maps, notes, diagrams) to extend thinking.

- **Cognitive offloading**: diaries, reminders, post-its, to-do lists reduce remembering effort
- **Computational offloading**: using paper/calculator to compute; representation matters (Roman numerals example)
- Annotation (underline/tick) + cognitive tracing (reordering items) are external support strategies

Design implication (MEMORIZE):
Provide external representations and visualizations that reduce memory load and help rapid decisions.

3.5 Embodied interaction

Embodied cognition says thinking is shaped by our body's engagement with the physical/social environment; we "think with our bodies," not just "through them."

Example: Touch gestures, VR interactions, tangible interfaces—movement changes understanding.

Summary (Week 4 in one paragraph)

Cognition includes attention, perception, memory, learning, and reasoning. Interface design affects how well people can perceive, focus, learn, and remember. Cognitive frameworks (mental models, gulfs, distributed, external, embodied cognition) help explain user behavior and guide design decisions.

WHAT TO MEMORIZE (Week 4 “Ratta List”)

1. Cognition definition + examples
2. Experiential vs reflective cognition (Norman)
3. Fast vs slow thinking (Kahneman) + examples
4. Attention: focused vs divided; Tullis results (5.5s vs 3.2s)
5. Driving + phone: hands-free not safer (cognitive distraction)
6. Perception design implications list (icons, spacing, contrast, sound, haptics)
7. Memory: working vs long-term; recognition vs recall (golden concept)
8. Misapplication of 7±2 in interface design
9. Learning: incidental vs intentional; learn by doing; exploration + guidance
10. Mental models definition + thermostat example + how to support good models
11. Gulfs of execution & evaluation definitions + what questions they answer
12. Distributed cognition definition + cockpit/ATC example
13. External cognition: cognitive offloading + computational offloading
14. Embodied interaction meaning

Practice Questions (with answers + clear explanations)

A) 5 MCQs

MCQ 1
Fast thinking (System 1) is best described as:
A) Slow, effortful, analytical
B) Automatic, intuitive, low effort
C) Only used for math problems
D) Always more accurate than slow thinking
Answer: B
Why: Kahneman's fast thinking is automatic and effortless (like 2+2).

MCQ 2
Tullis (1987) showed that search is faster when:
A) Information is bunched together
B) Information is grouped with spacing into categories
C) Information density is increased
D) Color contrast is the only grouping method
Answer: B
Why: Grouping + spacing reduced search time (3.2s vs 5.5s).

MCQ 3
Which statement about hands-free phone use while driving is correct (as per lecture)?
A) Hands-free is safe because both hands are on wheel
B) Hands-free is safer than passenger conversation
C) Hands-free is still dangerous due to cognitive distraction
D) It is safe if the road is empty
Answer: C
Why: Core issue is divided attention/cognitive load, not holding the phone.

MCQ 4
Humans are generally better at:
A) Recall than recognition
B) Recognition than recall
C) Neither, both are equal
D) Recall only when tired
Answer: B
Why: Interfaces should support recognition (menus/icons) rather than recall (commands).

MCQ 5
A mental model is:
A) The UI's visual design system
B) A user's internal understanding of how a system works
C) A database schema for app logic
D) A usability test script

Answer: B
Why: Mental model = internal understanding used to predict actions/outcomes.

B) 5 Short Answer Questions

SAQ 1: Define cognition and give 4 examples.
Answer: Cognition is mental processing for gaining understanding: thinking, remembering, learning, decision-making, reading, talking, writing, etc.
Explanation: Teacher wants definition + examples (not just the word).

SAQ 2: What is divided attention and why is it risky in interface use?
Answer: Divided attention is trying to attend to multiple tasks at once. It overloads focus capacity, causing errors and loss of train of thought; in driving it increases reaction time and accidents.
Explanation: A "busy" interface increases divided attention and error risk.

SAQ 3: Explain recognition vs recall using GUI vs command-line.
Answer: Recall requires retrieving from memory without cues (command-line: remember exact command). Recognition provides visible cues (GUI menus/icons) so user scans until they recognize. Humans are better at recognition.

SAQ 4: What are gulfs of execution and evaluation?
Answer: Gulf of execution is the gap between what user wants and what actions are possible/obvious ("what can I do/how?"). Gulf of evaluation is the gap between system state and user understanding ("what happened/did it work?"). Good design bridges both using clear actions + feedback.

SAQ 5: Describe the thermostat mental model error and what it teaches designers.
Answer: Many users think higher thermostat setting heats faster ("more is more"), but thermostat is on/off; higher setting doesn't change heating rate. Designers must support correct mental models via clear feedback, tutorials, and intuitive affordances.

C) 1 Case Study (with answer)

Case Study

Your university app has a "Fee Payment" flow. Students frequently:

- enter wrong info and fail payment,
- can't tell if payment went through,

- keep retrying buttons multiple times,
- get confused when switching between “Card / Bank Transfer / Wallet.”

Task: Use Week 4 cognition to explain the problems and propose fixes.

Model Answer

What’s going wrong (cognitive explanation):

1. **Attention overload:** Too many fields/messages at once causes scanning failures; key errors aren’t salient.
2. **Gulf of evaluation:** Users can’t interpret the system state (“Did it go through?”), so they repeat actions (like elevator button behavior).
3. **Poor mental model:** Users don’t understand the difference between payment methods or the steps required, so they make wrong inferences.
4. **Recall burden:** If the app makes them remember codes, steps, or where things are, it increases errors; recognition would help more.

Design fixes (concrete):

- **Make critical info salient**: highlight errors near the field, use spacing and grouping, reduce clutter.
- **Bridge gulf of execution**: clear step-by-step flow (“Step 1: Choose method → Step 2: Enter details → Step 3: Confirm”).
- **Bridge gulf of evaluation**: instant feedback (“Payment processing…”, “Payment successful”, receipt number, email sent).
- **Support recognition**: show “recent payment method,” prefill known info, use visible options and labels rather than hidden steps.
- **Support correct mental models**: short contextual explanations (“Wallet = instant, Bank Transfer = may take 1–2 hrs”).